

Agentic Environmental Data Proxy Discovery

A multi-agent pipeline that proposes, verifies, and catalogs alternative data sources for the Yale Environmental Performance Index

THE PROBLEM

EPI indicators are sparse & imputed

180 countries scored across 58 environmental indicators — but many rely on **modeled, not measured**, data.

- **Waste & recycling** — 56 countries imputed ($R^2=0.42$ vs GDP)
- **Wastewater** — single-year snapshots, sparse coverage
- **Pesticide risk** — only 2015 & 2018 globally

PARALLEL · COVID-19

Need: Official case counts under-reported — testing gaps, asymptomatic spread.

Mechanism: Infected people shed SARS-CoV-2 RNA in stool; sewage concentration tracks prevalence.

Ramifications: Wastewater signal forecast surges ~7 days ahead, independent of testing.

AGENT TOOLING

Data fetch & stats tools

30 Discovery

9 open data sources.

- World Bank
- WHO
- Comtrade
- NASA
- OpenAQ
- GEE

1 Stats Pkg

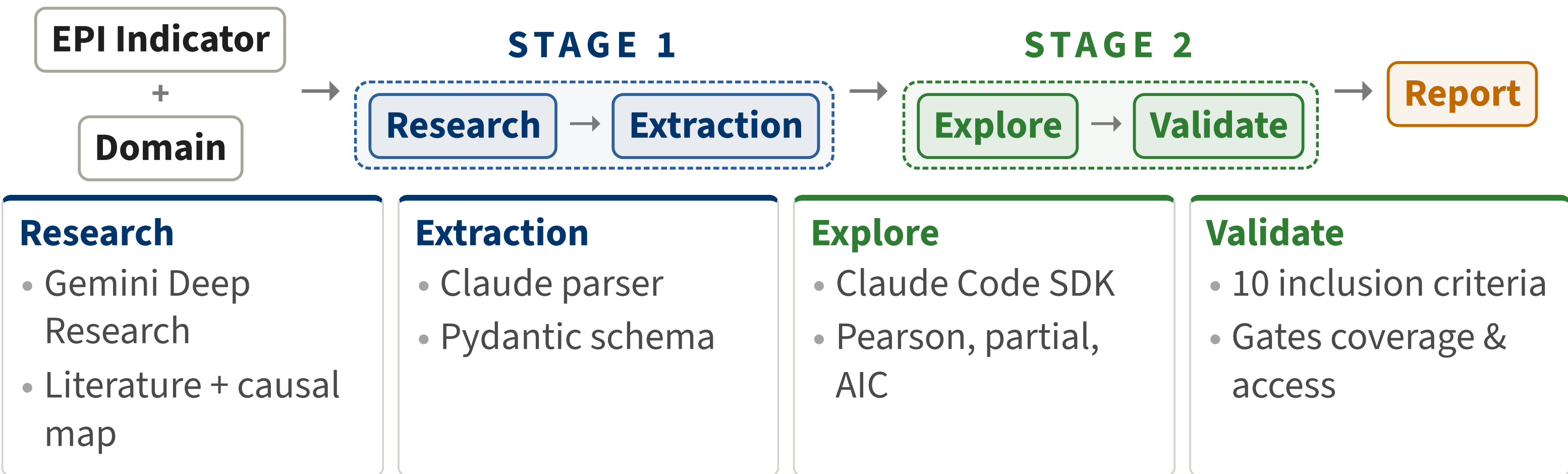
Verification utilities.

- Pearson
- Partial corr
- AIC fit
- Verdicts

METHOD

Multi-Agent Proxy Discovery Pipeline

Stage 1 drafts proxy hypotheses from the literature; Stage 2 verifies each with live data.



STEP 3 · KNOWLEDGE GRAPH COMPILATION

Compile: verified edges merge into one cross-indicator graph.

Transitivity: $A \leftrightarrow B$ and $B \leftrightarrow C$ surfaces $A \leftrightarrow C$ as a candidate.

Self-consistency: contradictory edge signs flag data issues.

Hub discovery: shared proxies link indicators (e.g., water access $\rightarrow$ UWD, PHL, SPI).

RESULTS

5 indicators · 48 hypotheses

48

Hypotheses

41

Confirmed / partial

85%

Inclusion rate

INDICATOR	STRONGEST PROXY	R
UWD	Basic water access	-0.81
UWD	Open defecation rate	+0.72
SPI	Wastewater per capita	+0.60
OEB	Obesity prevalence	-0.50

ARTIFACTS

Knowledge graph of verified proxies

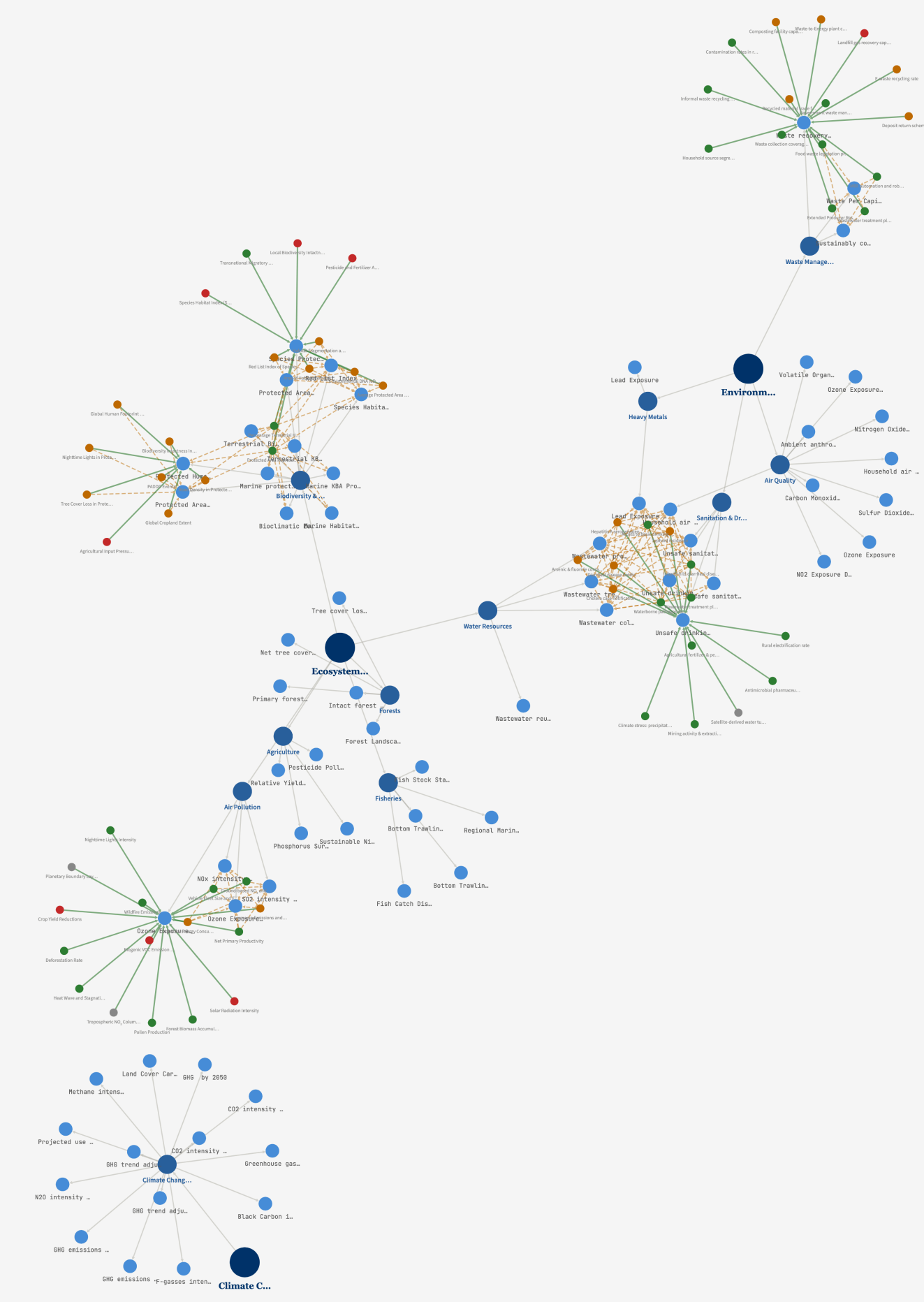


Fig 1. Cross-indicator graph (NetworkX + clingo ASP). Nodes are indicators and proxies; edges encode validated correlations.

Sample output — UWD dashboard

AUTO-GENERATED

UWD

Completed pipeline for UWD: 8 hypotheses, 8 verified

3 confirmed 4 partially confirmed 1 rejected

ID	PROXY VARIABLE	VERDICT	R	P	N	VALIDATION	INCLUSION
	RESEARCH AGENT	CODE AGENT	CODE AGENT	CODE AGENT	CODE AGENT	VALIDATOR AGENT	VALIDATOR AGENT
UWD-H01	Basic drinking water services access (% of population)	confirmed	-0.814	0.00e+00	4196	7 issues	9/10
UWD-H02	Open defecation rate (% of population)	confirmed	0.723	0.00e+00	4125	—	—
UWD-H03	Safely managed drinking water services (% of population)	confirmed	-0.697	0.00e+00	2987	—	—
UWD-H04	VIIRS Nighttime Light Radiance (mean)	partially confirmed	-0.233	2.19e-18	1372	9 issues	4/10
UWD-H05	Plastic pipes & fittings imports (HS 3917, USD)	partially confirmed	-0.239	1.80e-26	1938	—	—
UWD-H06	Pharmaceutical imports — retail medicaments (HS 3004, USD)	partially confirmed	-0.215	1.35e-21	1929	7 issues	6/10